Example: Sketch the graph of function $y = 2x^2 - 8x + 6$. As the coefficient of$x^2$ is positive, so the graph is $U$-*shaped* (Fig. 4.17)

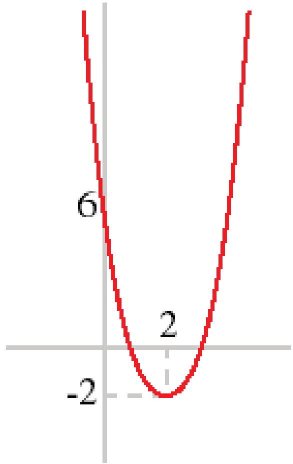


**Fig. 4.17:Graph of Function $y$=2x$^2$−8x+6**

In the graph of $y = x^2$, the point (0, 0) is called the vertex. The vertex is the lowest point in a parabola that opens upward. In a parabola that opens down ward, the vertex is the highest point. We can also graph a parabola with a different vertex. For example, see the graph of $y$=$x^2$+3 (Fig 4.18). The graph is shifted up 3 units from the graph of $y = x^2$, and the vertex is (0, 3).

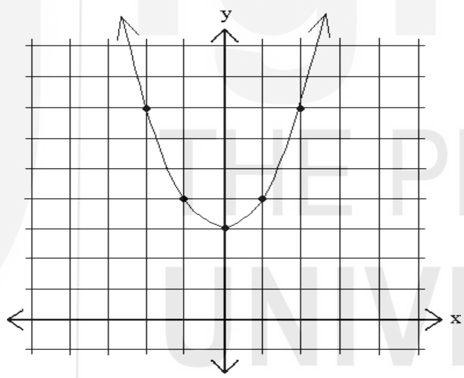


**Fig. 4.18: Graph of** $y$=x$^2$+3

We can also shift the vertex left and right. Observe the graph of $y$=($x$+3)$^2$ in Fig 4.19.

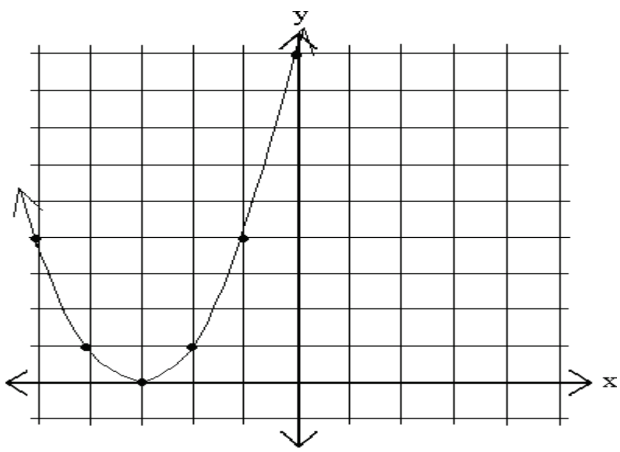


**Fig. 4.19: Graph of $y$=($x$+3)$^2$**

The graph is shifted *left* 3 units from the graph of $y$=$x^2$, and the vertex is (–3, 0). Next, see Fig. 4.20, the graph of $y$=($x$−3)$^2$, which is obtained by shifting to the right.

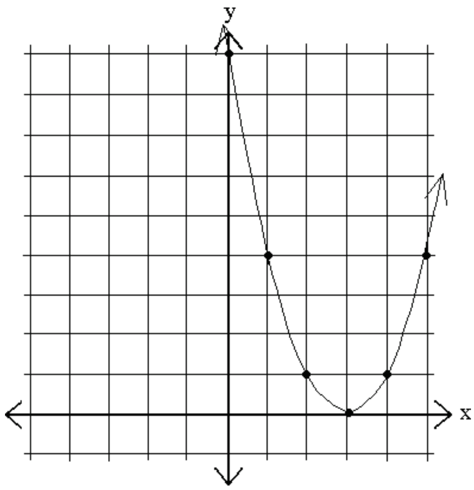


**Fig.4.20: Graph of $y$=($x$-3)$^2$**

The graph is shifted to the *right* 3 units from the graph of $y$=$x^2$, and the vertex is (3,0). Remember that the ***axis of symmetry*** is the line which divides the parabola into two symmetrical halves. It is given by the equation $x$=$h$. For example, the line of symmetry in the graph (Fig. 4.21) of $y$=($x$−2)$^2$+1is$x$=2.

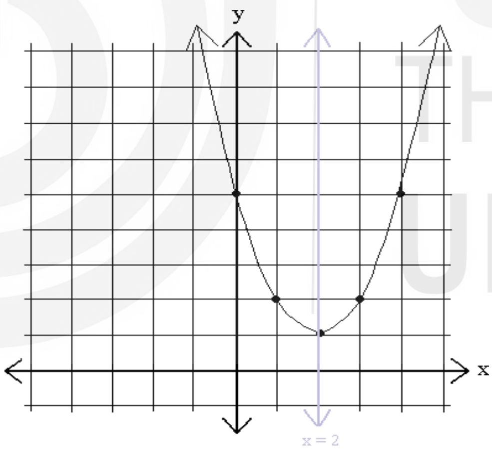


**Fig. 4.21: Axis of Symmetry**

### 4.4.2.1 Computation of the coordinates of Vertex

Consider a generic quadratic expression: $y$=$ax^2$+$bx$+$c$. Start with **completing the square** on the equation to have $y$=$a$ [$x^2$+$bx/a$+$c/a$] or $y$=$a$ [($x$+$b/2a$)$^2$−($b/2a$)$^2$+$c/a$]

The expression $-(b/2a)^2 + c/a - (b/2a)^2 + c/a$is a constant and it does not depend on $x$.So, we can replace it with $k$. Thus, we write$y$=$a$ [($x + b/2a$)$^2$+$k$]

Now, depending on whether $a$ is positive or negative, the parabola given by $y$ will either have either a maximum value or a minimum value. Since $a$ and $k$ are fixed, this must occur when $(x + b/2a)^2 = 0$. Hence, $x = -b/2a$ which implies that the function y has either a minimum value or a maximum value when$x = -b/2a$.

Thus, x-coordinate of the vertex is $-b/2a$. The y-coordinate can be obtained by replacing, in the equation $y=ax^2+bx+c$, the variable x by $(-b/2a)$.

**Example**: Find the vertex of $y=3x^2+x–2$ and graph the parabola.

To find the vertex, we look at the coefficients a, b and c. Then $k$ is obtained by evaluating $y$ at $h=–1/6$

x-coordinate of the vertex = $(-b/2a) = (-1/(2 \times 3)) = –1/6$.

The y-coordinate is given by replacing x by (–1/6) in the equation

$y=3x^2+x–2$.

$=3(– 1/6)^2+(– 1/6) –2=3/36–1/6 –2$

$=1/12–2/12 –24/12=–25/12$

We get y-coordinate as -25/12. Thus, the vertex is at (-1/6, -25/12). T-chart can be prepared as shown in Table 4.4.

**Table 4.4: T-chart of $y= 3x^2 +x - 2$**

| x | $y= 3x^2 +x – 2$ |
|---|---|
| -2 | 8 |
| -1 | 0 |
| 0 | -2 |
| 1 | 2 |
| 2 | 12 |

The graph (Fig. 4.22) can be drawn with the specification vertex:

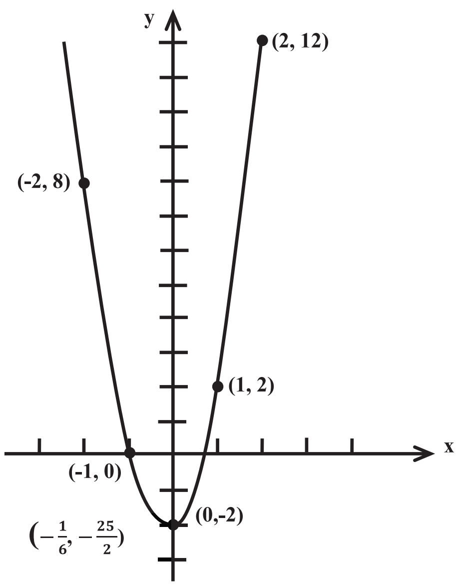


**Fig. 4.22: Graph of $y=3x^2+x-2$**

### 4.4.3 Cubic Functions

Cubic functions are encountered in the discussion of total cost in microeconomic analysis. Consider a cubic function of the form $f(x)= ax^3+bx^2+cx+d$. Its graph (Fig 4.23) is obtained as follows:

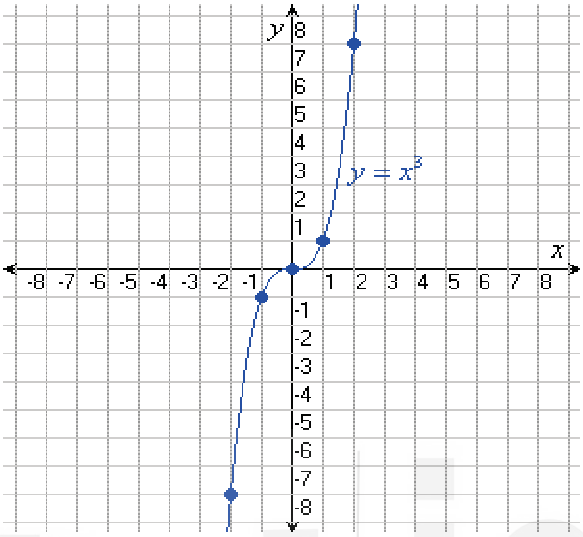


**Fig. 4.23: Graph of Cubic Function $f(x)=x^3$**

To trace the cubic function given by $f(x)=x^{3,}$ find the $x$ and $y$ intercepts of the graph of $f$, find the domain and range of $f$ and then sketch the graph of $f$. The y intercept is given by $(0, f(0))=(0, 0)$. The $x$ coordinates of the $x$-intercepts are the solutions to $x^3 = 0$. The only x-intercept is at the points (0,0). The domain of $f(x)$ is the set of all real numbers. Since the leading coefficient of $x^3$ is positive, the graph of f is up on the right and down on the left and hence the range of f is the set of all real numbers. Table of values is as follows:

| $x$ | -2 | -1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $f(x)=x^3$ | -8 | -1 | 0 | 1 | 8 |

Since f(-x) = - f(x), function f is odd and its graph is symmetric with respect to the origin (0,0).

## Check Your Progress 3

1) Draw the graph for the function, $y=x^2-2x$.

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2) What would be the shape of an odd powered Quadratic Function?

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## 4.5 GRAPHING ASYMPTOTIC FUNCTIONS

We are familiar with exponential growth models such as population growth and compound interest. Such models are represented by functions, which are called asymptotic functions. Next, we examine behavior of such functions through their graphical representations.

### 4.5.1 Graph of Asymptote

An asymptote is, essentially, a line that a graph approaches, but does not intersect (Fig. 4.24). For example, in the following graph of *y=1/x*, the line approaches the x-axis (y=0), but never touches it. No matter how far we go into infinity, the line will not actually reach y=0, but will always get closer and closer to it.

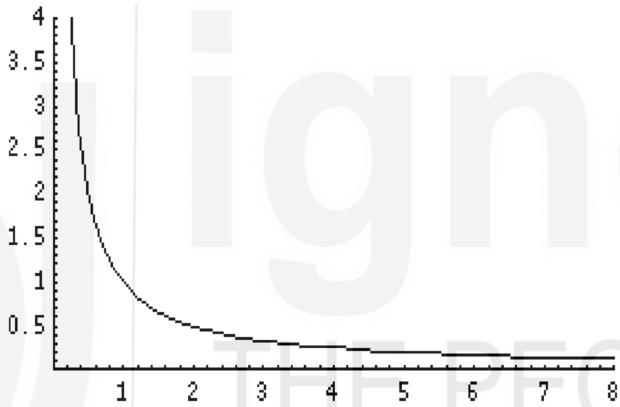


**Fig. 4.24: Graph of Asymptote**

***This means that the line y=0 is a horizontal asymptote***. Horizontal asymptotes occur most often when the function is a fraction where the top remains positive, but the bottom goes to infinity. Going back to the previous example, y=1/x is a fraction: When we go out to infinity on the x-axis, the top of the fraction remains 1, but the bottom gets bigger and bigger. As a result, the entire fraction gets smaller, although it will not hit zero. As we increase the values of x, the values of the function will gradually have values 1/2, then 1/3, then 1/10, even 1/10000, but never quite 0. Thus, y=0is a horizontal asymptote for ***the function y=1/x.***

The function also serves as an example of ***vertical asymptote, because as y becomes larger and larger in absolute value, x becomes smaller and smaller in magnitude, but never becomes 0.***

Similarly, the equation y =4/(x−2), gives one of the horizontal asymptotes, viz. y= 0, and one of the vertical asymptotes, viz. x=2 (Fig. 4.25).

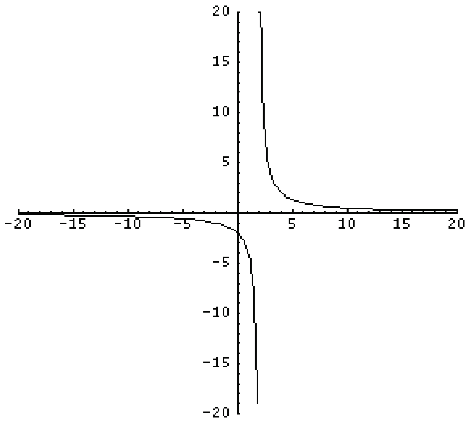


**Fig. 4.25: Graph of the Function y =4/(x−2)**

Note that as the graph approaches x=2 from the left, the curve drops rapidly towards negative infinity. This is because the **numerator** is staying at 4, and the denominator is getting close to 0. That means that the fraction itself is getting very big and negative. When x is exactly 2, the function does not exist because you cannot divide by 0. Immediately after 2 it resumes at positive infinity, because the numerator is 4 and the denominator is again very tiny, but this time it is positive.

Next, we consider two examples of functions one of which gives only (i) horizontal asymptote, but not a vertical asymptote, and (ii) vertical asymptote, which is not horizontal.

i) The function y= 1/ (x –2), for $0 \leq x < 2$, is asymptotic with x= 2, being a vertical asymptote. Similarly,

ii) The function x= 1/ (y –2), for $0 \leq y < 2$ is asymptotic with y= 2, being a horizontal asymptote

## 4.5.2 Square Root Function

As the name indicates we take a variable x to write the function in the form off(x)=√x, with Domain/Range: [0,∞), and intercept (0,0), and which is neither even nor odd and is increasing on(0,∞). Plotting its graph yields Fig. 4.26.

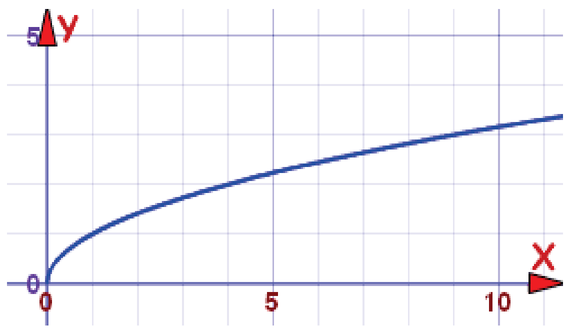


**Fig. 4.26: Graph of Square-root Function**

You may see that square root function is used to depict an *is o cline* which provides information on the way different is o quant are related to each other. In fact, this is o cline based on square root function converges to a point where output is maximized and marginal product is zero.

### 4.5.3 Exponential Function

A general exponential function is of form y =$a^x$, where a is an arbitrary constant, and x any real variable. The discussion of the general exponential function requires substantial space, and hence, will not be discussed here. We discuss a special such function which is found useful in several applications, viz. the natural exponential, which is of the for my= $e^x$where $e$ is Euler's number (a number approximately equal to 2.718281828). If we take its domain as the set of all real numbers i.e., $(-\infty, \infty)$, then its range is $(0, \infty)$. Since the graph never intersects or goes below the x-axis, the y-values are not zero or negative (Fig. 4.27).

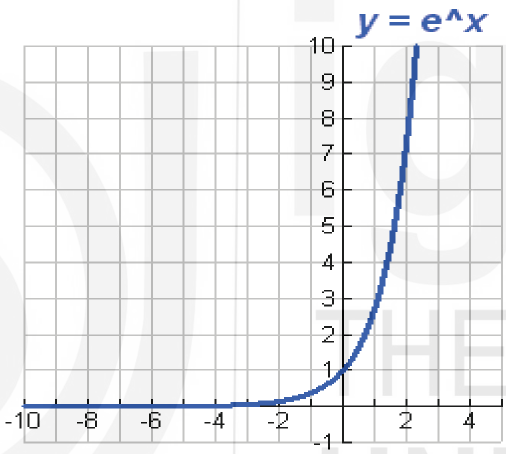


**Fig. 4.27: Graph of Exponential Function**

Three of the most common applications of exponential and logarithmic functions (to be discussed soon) have to do with interest earned on an investment, population growth, and carbon dating. For example, exponential functions are used to describe growth or decay processes with constant growth (or decay rate), population growth, growth of GDP, inflation etc.

### 4.5.4 Logarithmic Function

A logarithm function is a sort of inverse function of exponential function. Thus, if y = $b^x$=f(x) is an exponential function, then x = $\log_b$ y = g(y) (read as log of y to the base b) is such that, with some restrictions, f(g(y)) = y, and g(f(x)) = x. A function given by $y=log_bx$**,** where $b$ is any number such that $b$>0, $b$≠1, *and* $x$> 0, denotes logarithmic function. When $b$> 1, the values of the logarithm increase; and for 0<$b$< 1, the values decrease.

The logarithmic function slowly grows towards positive infinity as the variable increases and slowly goes towards negative infinity as it approaches 0. Three special cases of logarithm are quite well-known and useful: (i) b =10, called decimal or common logarithm and is more used in science and engineering applications. (ii) b = e, approximately equal to 2.718281828 is called natural logarithm and used frequently in Mathematics and physics, and (iii) base =2, is called b in arylogarithm, and is frequently used in computer science. For all bases b, including the three mentioned here, $\log_b 1 = 0$

In the logarithmic function, the domain is all positive real numbers (never zero) while the range is all real numbers. The graph is asymptotic to the *y*-axis – gets very, very close to the *y*-axis but does not touch it or cross it (Fig 4.28).

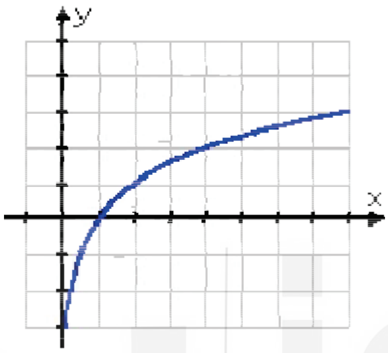


**Fig. 4.28: Graph of Logarithmic Function**

## Check Your Progress 4

1) Let *f* be a rational function defined by $f(x)=x+1/x-1$.Find the x and y intercepts of the graphs of the function.

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2) Draw the graph of an average fixed cost curve. Why do you think it is asymptomatic?

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## 4.6 GRAPHING PIECEWISE FUNCTIONS

Up to now, in this chapter, a function has been represented by a single equation. In many real-life problems, however, functions are represented by a combination of equations, each corresponding to a part of the domain. Such functions are called ***piece wise functions***, graphs of which will be discussed below.

### 4.6.1 Rational Function

A rational function can be represented as the quotient of polynomials, just as a rational number is a number which can be expressed as a quotient of whole numbers. The graphs of rational functions can be very complicated sporting bumps, holes and asymptotes. Consider a rational function that is a quotient of polynomials g(x) and $h(x) \neq 0$.

It looks like

$y = g(x)/h(x)$

Where $g$ and $h$ are polynomials ($h \neq 0$). $f(x)=x+2/\ x-5)$ is a rational function.

It is not necessary that the denominator of the function must have variable to be in this category of function. Note that the domains of the rational functions must be restricted to values of x that will not make the denominator of the function equal to zero.

**Table 4.5: T-chart Value**

| X | f(x) |
|---|---|
| -4 | -.25 |
| -2 | -0.5 |
| -1 | -1 |
| -.1 | -10 |
| -.01 | - 100 |
| .01 | 10 |
| 1 | 1 |
| 2 | 0.5 |
| 4 | 0.25 |

The range of a rational function is sometimes easier to find by first finding the inverse of the function and determining its domain. Let us sketch the graph of $(x)=1/x$. For drawing the graph, we must be careful that for no value of the independent variable, denominator becomes 0. Now, by plugging in some values of $x$ and we get Table 4.5.

So, as $x$ gets larger in magnitude (positively and negatively) the functional value has the same sign as the sign of $x$ *but* reduces in magnitude. Likewise, as we approach $x = 0$ the function again keeps the same sign as $x$ but starts getting quite larger. It gets very close to the $y$-axis but will never cross or touch it. As a result, there will be no $y$-intercept. Here is the sketch of this graph (Fig. 4.29).

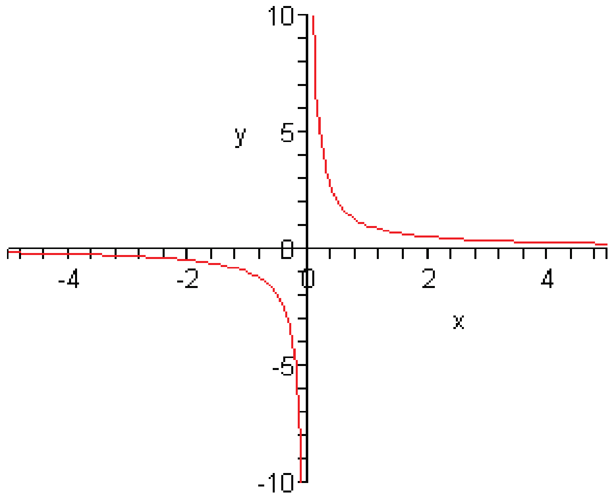


**Fig. 4.29: Graph of Rational Function**

You may see that the graph is in two pieces. Most of the rational functions will have graphs in multiple pieces. As may be seen the graph does not have any intercepts (with any of the axes) of any kind. Similarly, we get $x$-intercepts by solving $f(x) = 0$. In the example above the numerator is one and will never be zero. So, the function will haven o$x$-intercepts. Again, the graph will get very close to the $x$-axis but it will never touch or cross it. In the graph above (Fig. 4.29) as the value of $x$approaches $x = 0$ the graph starts gets very large on both sides of the line given by $x = 0$. This line is called a **vertical asymptote.** Also, as $x$get very large, both positive and negative, the graph approaches the line given by $y$=0.This line is called a **horizontal asymptote**.

### 4.6.2 Piecewise Functions

Piecewise functions (split functions) are defined in pieces. So, their graphs are drawn in pieces. For instance, suppose we have $y = \begin{cases} x^2 - 2, x<1 \\ -2x + 4, \geq 1 \end{cases}$

Since this function has two pieces, you may find it helpful to work with two sets of $x$ and $y$ values independently, one at a time; if it had more pieces, work with each piece independently and draw graph for each, one at a time more charts. In this case, the break between the two "halves" of the function (the point at which the function changes rules) is at $x$=1, so that is where your charts will break (Table 4.5).

**Table 4.5:Chart of the piecewise function**

| $x$ | $y=x^2-2$ | $x$ | $y=-2x+4$ |
|---|---|---|---|
| -4 | 14 | 1 | 2 |
| -3 | 7 | 2 | 0 |
| -2 | 2 | 3 | -2 |
| -1 | -1 | 4 | -4 |
| 0 | -2 | | |
| 1 | -1 | | |

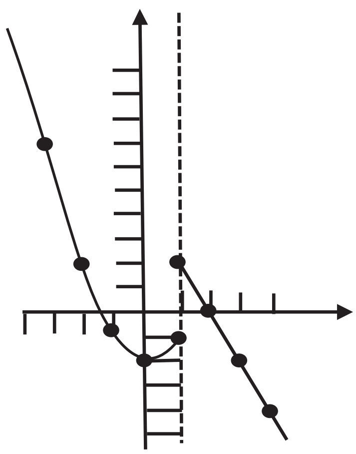

**Fig. 4.30: Graph of the Piecewise Function**